

Tugas Week 11
Jaringan Komputer



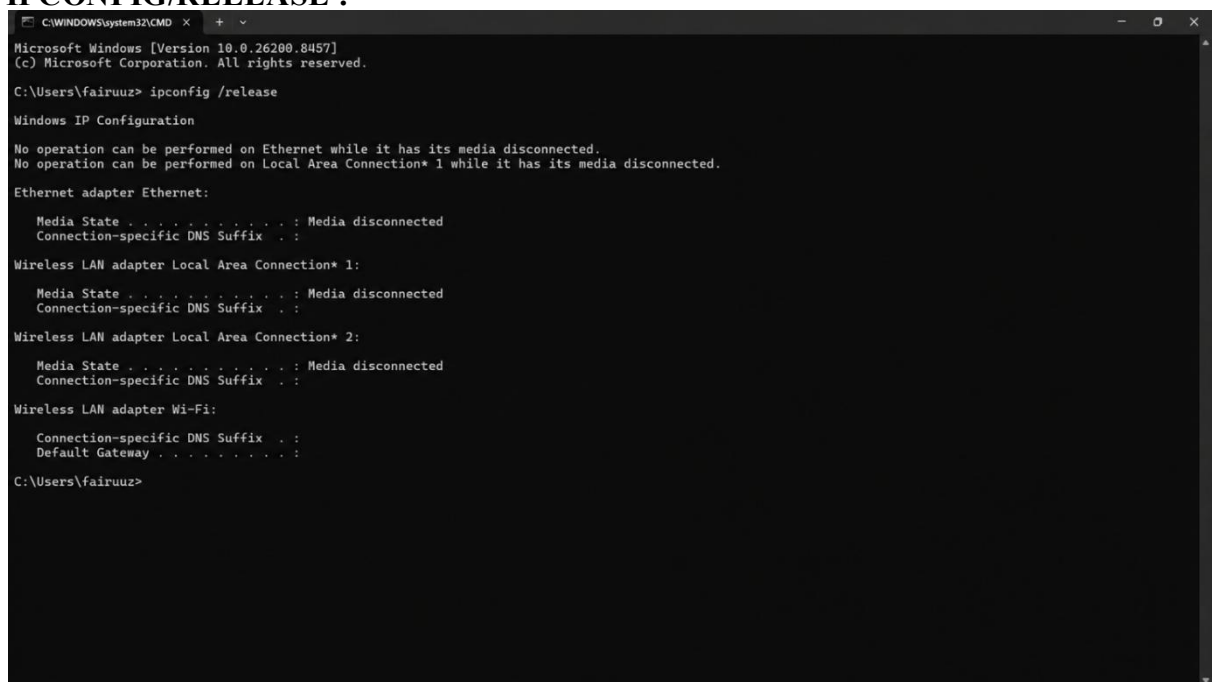
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1. Mahasiswa dapat menginvestigasi cara kerja protokol DHCP menggunakan Wireshark

Di modul ini, kita akan melihat sekilas Dynamic Host Configuration Protocol, DHCP. Ingat bahwa DHCP digunakan secara luas di perusahaan, universitas dan LAN kabel dan nirkabel jaringan rumah untuk secara dinamis menetapkan alamat IP ke host, serta untuk mengkonfigurasi informasi konfigurasi jaringan lainnya. Seperti yang telah kami lakukan di lab Wireshark sebelumnya, Anda akan melakukan beberapa tindakan di komputer Anda yang akan menyebabkan DHCP beraksi, dan kemudian menggunakan Wireshark untuk mengumpulkan dan kemudian jejak paket yang berisi pesan protokol DHCP.

Mengumpulkan Jejak Paket :

1. IPCONFIG/RELEASE :



```
C:\WINDOWS\system32\CMD
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fairuuz> ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Default Gateway . . . . . :

C:\Users\fairuuz>
```

2. IPCONFIG/RENEW :

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Wi-Fi while it has its media disconnected.

C:\Windows\System32>ipconfig /renew

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . : fe80::3932:75d0:5e6f:8c31%12
    IPv4 Address. . . . . : 192.168.1.53
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Windows\System32>
```

3. DHCP REQUEST :

The image shows a Wireshark packet capture of a DHCP request. The packet list shows a DHCP request from 192.168.1.53 to 192.168.1.1. The packet details show the DHCP request structure, including the message type (Boot Request), hardware type (Ethernet), and transaction ID (0xb26ae3d3). The packet bytes show the raw data of the DHCP request.

No.	Time	Source	Destination	Protocol	Length	Info
2	0.029406400	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xb195949f
79	2.595543800	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xb195949f
95	2.747688400	192.168.1.1	255.255.255.255	DHCP	357	DHCP ACK - Transaction ID 0xb195949f
1223	19.566603800	192.168.1.53	192.168.1.1	DHCP	358	DHCP Request - Transaction ID 0xb26ae3d3
1224	19.610818500	192.168.1.1	192.168.1.53	DHCP	357	DHCP ACK - Transaction ID 0xb26ae3d3

Frame 1223: Packet, 358 bytes on wire (2864 bits), 358 bytes captured (2864 bits) on interface...

Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: zte_23:db:9a (bc:bd:84:23:db:9a)

Internet Protocol Version 4, Src: 192.168.1.53, Dst: 192.168.1.1

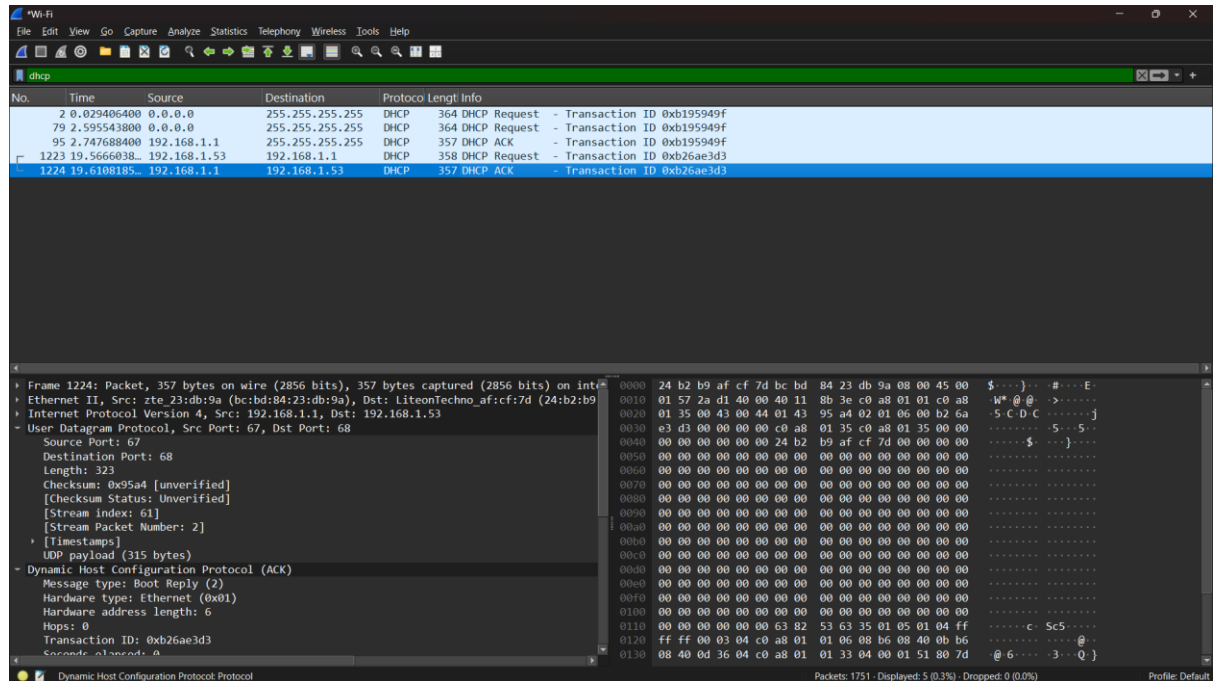
User Datagram Protocol, Src Port: 68, Dst Port: 67

Source Port: 68
Destination Port: 67
Length: 324
Checksum: 0x4f0f [unverified]
[Checksum Status: Unverified]
[Stream index: 61]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (316 bytes)

Dynamic Host Configuration Protocol (Request)

Message type: Boot Request (1)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0xb26ae3d3
Secondary transaction ID: 0

4. DHCPACK



Pada percobaan DHCP di Windows, paket yang tertangkap hanya DHCP Request dan DHCP ACK. Hal ini terjadi karena sistem masih memiliki lease IP sebelumnya sehingga proses DHCP berlangsung dalam mode renewal, bukan initial discovery penuh.